



Evaluating the effects of alternative forest management strategies on spruce budworm outbreaks

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Received: 9 May 2025 / Accepted: 9 October 2025 / Published online: 23 November 2025
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Abstract

Context Forest insect outbreaks are major landscape-scale disturbances that shape forest composition and dynamics. In eastern North America, the spruce budworm (SBW) is a key defoliator driving these patterns. Understanding how local forest management interacts with outbreak dynamics at broad spatial extents is crucial for designing sustainable management strategies under climate change.

Objective We evaluated whether forest management strategies can modify SBW outbreak development across broad spatial extents and over long time frames.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10980-025-02242-6>.

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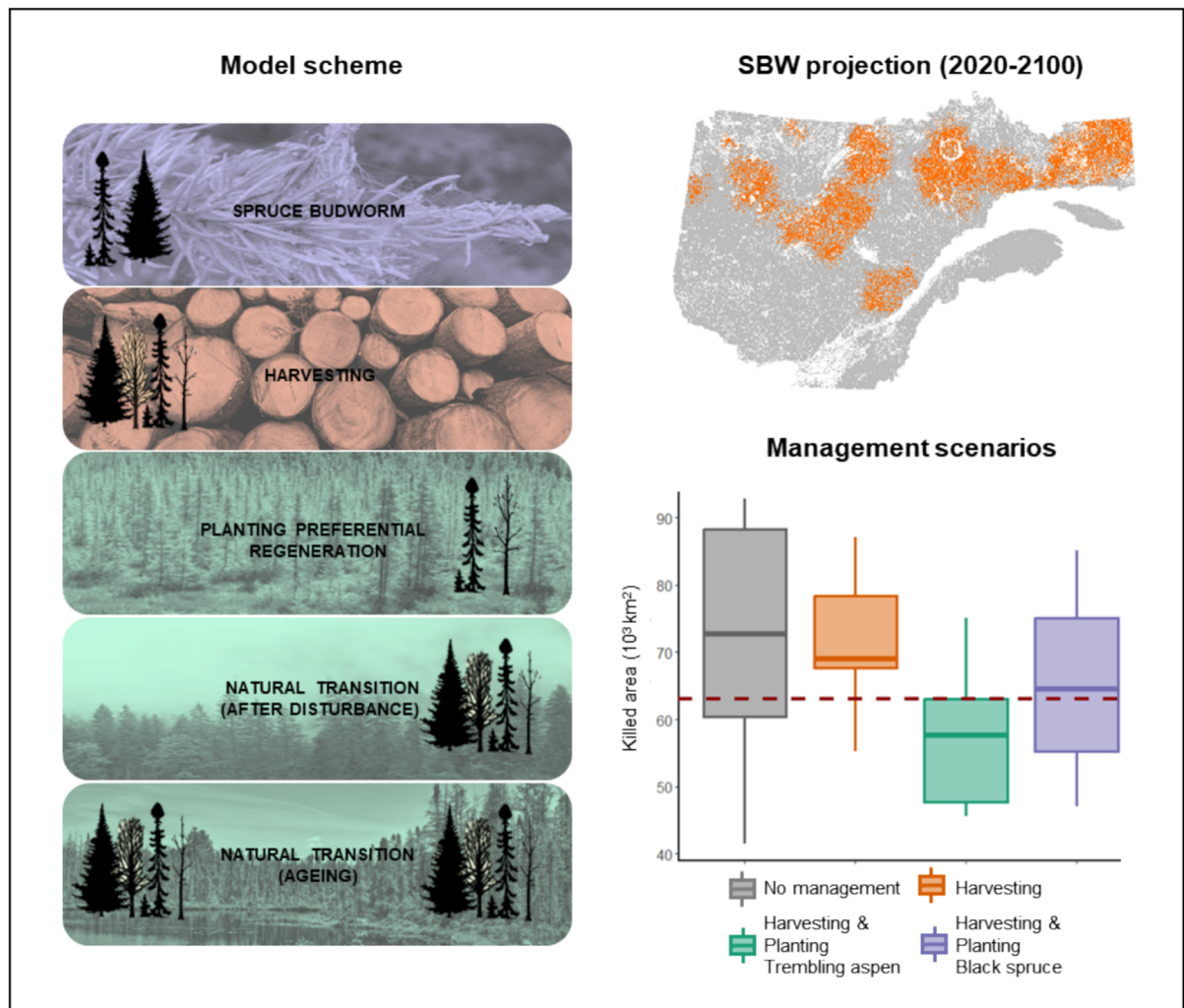
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Methods We developed a spatially explicit forest landscape model to simulate SBW outbreaks in Quebec, Canada, over 80 years. Simulations were used to compare scenarios with different harvesting rates and regeneration strategies under current and future climate conditions.

Results Outbreaks affected from 60,000 to 160,000 km² (11–31% of forest area), with climate change driving a gradual northward shift in maximum outbreak impacts. Annual harvesting of 1% of the forest and promoting non-host regeneration (e.g., trembling aspen) reduced SBW-induced forest mortality by up to 30%, particularly when combined with higher harvest intensities that accelerated species turnover and advanced dampening effects by one to two decades. In contrast, prioritizing host species (e.g., black spruce) had little effect on outbreak patterns.

Conclusions Local management strategies can substantially influence large-extent outbreak dynamics if applied consistently and intensively. However, our results highlight that even proactive measures will only partially offset future risks, as climate-driven northward shifts may reduce management effectiveness. Spatially explicit models provide valuable insights to design adaptive forest management that mitigates insect disturbances under climate change.

Graphical Abstract



Keywords Landscape simulation model · Insect outbreaks · Harvesting · Plantation-based regeneration · Silviculture · Spruce budworm

Introduction

Insects are key agents of change, shaping forest landscape composition and structure and altering forest ecosystem functioning (Hulme 2009; Turner 2010). Forest insect outbreaks can cause significant impacts on ecosystem services, affecting not only the forest industry and the economic viability of communities

that depend on it, but also non-market values such as recreation, aesthetics, biodiversity conservation and homeowner property benefits (Chang et al. 2012; Boyd et al. 2013; Niquidet et al. 2016; Freer-Smith and Webber 2017). Such impacts are increasing due to global change, as new climatic conditions alter both insect development and population dynamics (Jactel et al. 2019; Lehmann et al. 2020) and increase host plants susceptibility (Jaworski and Hilszczański 2014). Recent decades have witnessed some of the most severe insect outbreaks ever recorded. For example, the unprecedented mountain pine beetle (*Dendroctonus ponderosae*) outbreak in western

North America affected over 18 million hectares of forest and caused economic losses in the billions of dollars (Kurz et al. 2008). Similarly, the European spruce bark beetle (*Ips typographus*) has damaged over 40 million m³ of timber since 2018 (Hlásny et al. 2021). Together with the current spruce budworm outbreak in Eastern Canada that has extended farther north than previous recorded outbreaks, these cases illustrate how global change is altering forest insect outbreaks worldwide.

A variety of management or population control strategies have been explored to mitigate the risks posed by insect pests. Forest management practices that alter forest composition and structure through host tree removal or promoting the presence of non-host species and natural enemies can reduce forest vulnerability to outbreaks (Klapwijk et al. 2016; Marini et al. 2022). Aerial insecticide application, mass trapping, sanitary cutting of infested trees, and preemptive harvesting of susceptible hosts have also been used to slow outbreak spread (Régnière et al. 2019a, b; MacLean 2019). In Eastern Canada, for instance, the spruce budworm early intervention strategy targets incipient population buildups through localized insecticide treatments, aiming to delay or prevent full-scale outbreaks (Johns et al. 2019). While promising, these potential strategies often face limitations related to costs, logistics, and long-term effectiveness (Johns et al. 2019).

The spruce budworm (*Choristoneura fumiferana* Clem. (SBW)) is the most damaging insect in Eastern North American coniferous forests (MacLean 2004; Pothier et al. 2012). SBW primarily defoliates and causes mortality on balsam fir (*Abies balsamea* (L.) Mill.), but also on spruce species such as black spruce (*Picea mariana* (Mill.) B.S.P.), white spruce (*P. glauca* (Moench) Voss), and red spruce (*P. rubens* Sarg.). In the region, large-extent synchronous outbreaks typically occurred every 30–35 years during the last centuries, showing a relatively cyclic temporal pattern combining endemic, pre-epidemic, epidemic, and collapse phases (Blais 1983; Royama 1984; Fleming 2000; Chabot et al. 2014; Cooke 2024). The SBW is a lepidopteran that is primarily dispersed by flying, and wind patterns have been shown to facilitate its spread across large distances, contributing to outbreak expansion (Régnière et al. 2019a, b; Gandiaga and James 2023). Climate conditions also play a crucial role in SBW population dynamics, affecting

insect development rates, fecundity, dispersal and migration movement, survival, as well as the development and efficiency of SBW parasitoids (Candau and Fleming 2011; Gray 2013). Temperature was identified as the main limiting factor for SBW expansion in terms of latitude and elevation (Régnière et al. 2012; Boakye et al. 2022). However, the effects of climate change on SBW dynamics are not always straightforward. While warmer temperatures tend to accelerate larval development and expand the insect's potential range, seasonal mismatches—particularly warming in autumn, which can decouple larval development from host phenology or delay entry into diapause—can negatively affect SBW overwintering survival and fecundity (Régnière et al. 2012). This suggests that both the timing and the magnitude of warming matter, and that non-linear responses can occur depending on regional climatic regimes. Additionally, climate affects not only the physiology and phenology of SBW itself, but also the synchrony with its host trees and the effectiveness of its natural enemies, such as parasitoids, whose development rates are also temperature-dependent (Candau and Fleming 2011).

Forest characteristics were also identified as important factors in SBW dynamics, including factors such as stand structure and composition (Bouchard and Auger 2014), age of host species (Erdle and MacLean 1999), and site quality for the host species (e.g. climatic and soil conditions; Fuentealba and Bauce 2012). Large concentrations of mature stands of host tree species are also important for the location of SBW outbreak epicenters—outbreak starting points—and their expansion (Blais 1983). Through their influence on forest composition and structure, forest management practices have been historically theorized to shape SBW dynamics in terms of frequency, synchrony, expansion, and intensity (Miller and Rusnok 1993; Robert et al. 2012). In defoliators systems, this effect has been attributed to the direct effects of within stand dispersal mortality (Zhang et al. 2020) and the indirect effects of non-host abundance and diversity on SBW parasitism rates (Eveleigh et al. 2007; Legault et al. 2018). Similar patterns have been observed in bark beetles effects in Europe, where long-term management favoring Norway spruce has contributed to widespread outbreaks of the bark beetle *Ips typographus* under recent climate stress (Wermelinger 2004; Raffa et al. 2008; Kozhoridze et al. 2024). In this context,

forest management strategies that promote a greater abundance of non-host species have been suggested to reduce forest vulnerability to insect outbreaks at the landscape level (Miller and Rusnock 1993; Pureswaran et al. 2016; Kozhoridze et al. 2024). In the case of the spruce budworm, increasing the proportion of less preferred hosts such as black spruce, or of non-host deciduous species such as trembling aspen or white birch (*Betula papyrifera*), have already been considered as potential measures to limit outbreak impacts (Miller and Rusnock, 1993; Sainte-Marie et al. 2015). Still, the potential influence of this type of measure on outbreak development at larger spatial extents has not been examined.

The study of large-extent and long-term patterns in insect outbreaks and their relationships with forest composition is challenging due to the multiple potential interactions between landscape configuration, forest dynamics, forest management, insect population dynamics, and climate. In this context, spatially explicit forest landscape models are useful tools to investigate the multi-decadal dynamics resulting from the interactions between forest succession processes and insect disturbances under future global change scenarios (Keane et al. 2004; 2015; Gustafson et al. 2010; Rammer 2024). For example, simulation approaches have been applied to other major forest pests, including *Dendroctonus ponderosae* in Western North America and *Ips typographus* in Central Europe, to assess outbreak risk and evaluate management responses under climate change (Temperli et al. 2013; Seidl et al. 2017). Forest landscape models are particularly relevant to study the impact of the SBW because of its long population cycles (usually 30 years or more), and the extensive patterns of forest defoliation and mortality that it causes (Robert et al. 2020). Furthermore, such models allow testing the long-term and wide-extent impact of alternative forest management scenarios, which interact with forest succession and natural disturbances across different scales (James et al. 2011; Bouchard et al. 2019).

Evaluating the potential of different forest management strategies to reduce the impact of SBW outbreaks is essential given its major ecological and economic implications in eastern Canada. In the province of Quebec, the area affected by the current (2006) outbreak cycle reached more than 12 million ha in 2021 alone, requiring intensive protection efforts that exceeded 50 million Canadian dollars for

that year (QMFFP 2022). In this study, we developed a spatially explicit landscape simulation model that integrates SBW dynamics, forest succession, and forest management across a very large area overlapping different biomes. These processes were modeled as climate-sensitive, allowing simulations to account for forest and insect responses to future environmental conditions. Our study aims to (1) determine how climate change affects the spatial distribution and duration of SBW defoliation as well as forest mortality in response to different climatic scenarios, and (2) evaluate whether modulating harvesting levels and type of reforestation could be effective in mitigating SBW impacts under future climate conditions.

Materials and methods

Study area

The study area covers 589,856 km² in the southern portion of Quebec, Canada, across a climatic gradient ranging from ~0 °C mean annual temperature in the boreal north to ~6 °C in the temperate south, with mean annual precipitation between 800 and 1200 mm (Subedi et al. 2023). Vegetation is structured into three ecozones from south to north: temperate, mixed, and boreal (Olson et al. 2001). The boreal zone is dominated by conifer species such as black spruce (*Picea mariana*) and balsam fir (*Abies balsamea*), with some shade-intolerant hardwoods like trembling aspen (*Populus tremuloides*) and white birch (*Betula papyrifera*; Fig. 1). The temperate zone is richer in species diversity and it is dominated by sugar maple (*Acer saccharum*). The mixed ecozone contains a mixture of boreal and temperate species. Forest management is widespread: approximately 65% of the landscape is designated for timber production, structured under Forest Management Units (FMUs) defined in tactical integrated forest management plans (QMFFP 2018). Harvesting primarily relies on partial cuts in the temperate and mixed zones and clearcuts or shelterwood systems in the boreal regions. Protected areas of various sizes are distributed across the territory, and the northern portion, though forested, is considered less productive and remains unmanaged due to remoteness and poor accessibility (Bouchard 2023). The main natural disturbances impacting this area include SBW outbreaks, wildfires, and

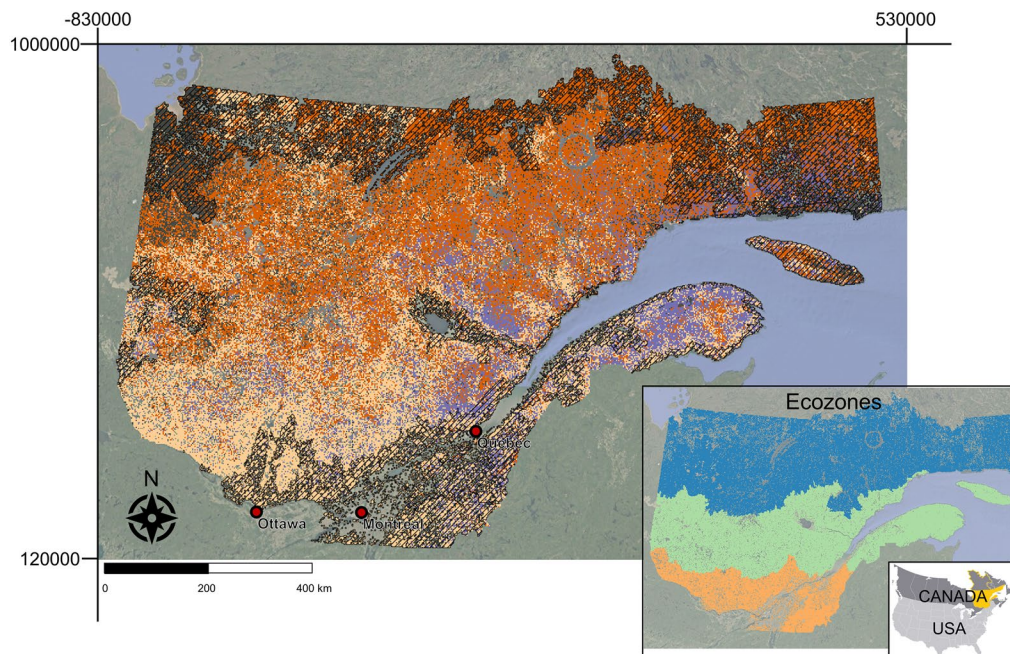


Fig. 1 Study area. Map of the study area at a resolution of 4 km² representing the distribution of species affected by the SBW in 2020: black spruce (*Picea mariana*) depicted in orange, balsam fir (*Abies balsamea*) in purple, other forest species in cream, absence of forest in gray and striped lines represent the area beyond the northern limit of forest management

activities (to the north) and private lands (to the south of the province). Ecozones of the study area are included: boreal forest is represented in blue, mixed forests in green and temperate forests in orange. Extension of study area is incorporated (in meters, EPSG:2154 projection). Source: <https://www.foret-ouverte.gouv.qc.ca/>, accessed in December 2020.

windthrow, with SBW being particularly impactful over the course of the last century (Pureswaran et al. 2016). Major outbreaks occurred in the 1910s, 1940s, 1970s, and the current one has already affected over 10 million ha, especially in balsam fir stands (Boulangier and Arseneault 2004).

Model overview

We developed a spatially explicit landscape model to simulate the interactions between spruce budworm outbreaks development, forest management activities, and forest dynamics (Fig. 2). This model is derived from the Quebec Landscape Dynamic Model, which has been previously used to study forest responses to climate change (Bouchard et al. 2019) and the interaction between fire and harvesting (Bouchard et al. 2023). In this version, we incorporated the representation of SBW dynamics and impacts. The model operates on a raster grid at 4 km² resolution across the entire study area and was simulated with annual

time steps over the period 2020–2100. It includes four main processes: (1) large-extent SBW outbreak development across endemic, pre-epidemic, epidemic, and collapse phases, affecting forest stands through defoliation and mortality; (2) wood harvesting of mature stands; (3) preferential regeneration of non-host (trembling aspen) or secondary host species (black spruce) in harvested cells; and (4) forest dynamics (e.g., stand aging, natural mortality, and composition shifts). The model was developed in R (R Core Team 2022) and is available on GitHub (<https://github.com/quimcanelles/SBW-MGMT>).

State variables and initialization

Each cell in the model is characterized by different types of state variables: compositional (cell age and dominant forest species), environmental (elevation, temperature, precipitation and soil moisture), defoliation status, and forest management unit. Initialization data were prepared at a 4 km² resolution for the year

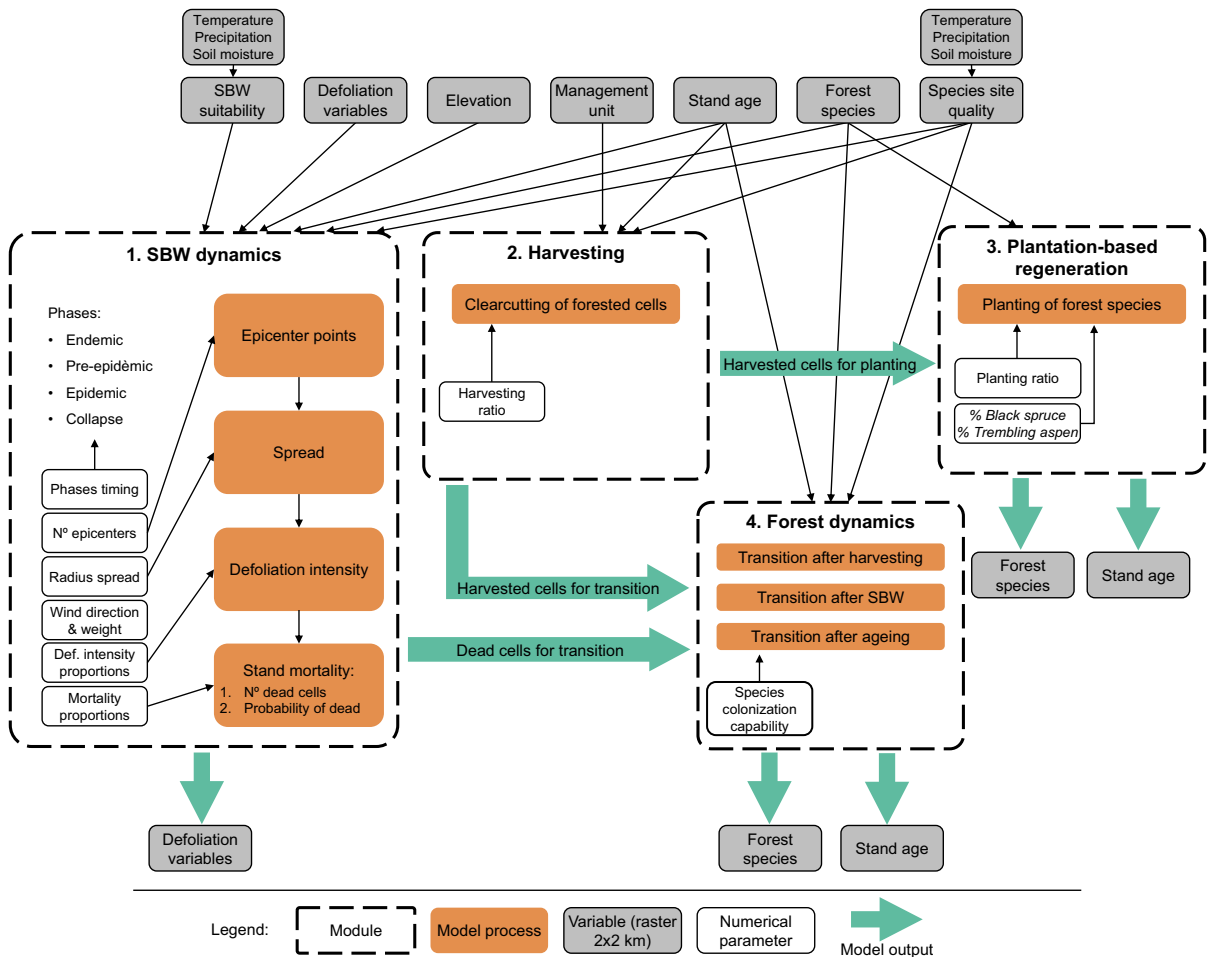


Fig. 2 Model scheme. Main processes simulated in the modeling framework. This model integrates four core modules: SBW dynamics, harvesting, plantation-based regeneration, and

Forest dynamics. The figure illustrates the essential processes, variables, and parameters required for simulating each module effectively

2020, marking the initial year of model simulations. This section describes the state variables and how they were initialized for the study area.

Composition and forest variables

The compositional variables in each 4 km² cell x and time step t correspond to its dominant forest species ($spp_{x,t}$) and its age ($age_{x,t}$), defined as time since the last stand-replacing disturbance (harvesting or SBW-induced mortality). We considered five possible dominant tree species: sugar maple (*Acer saccharum*), yellow birch (*Betula alleghaniensis*), trembling aspen (*Populus tremuloides*), balsam fir (*Abies balsamea*), and black spruce (*Picea mariana*). Less common

species from boreal forests like white birch (*Betula papyrifera*), jack pine (*Pinus banksiana*), tamarack (*Larix laricina*), and white spruce (*Picea glauca*) were grouped under the “Other boreal forest” category. Similarly, the “Other temperate forest” category included basswood (*Tilia americana*), beech (*Fagus grandifolia*), oaks (*Quercus* spp.), red maple (*Acer rubrum*), white pine (*Pinus strobus*), red pine (*Pinus resinosa*), eastern hemlock (*Tsuga canadensis*), and red spruce (*Picea rubens*). Initial forest characteristics were obtained from the forest maps in shapefile format of the Ministère des Ressources naturelles et des Forêts du Québec (QMFFP 2023), and used to assess such tree species composition and stand age

for each cell by identifying the composition of the stand located at its center.

Defoliation variables

We assumed that defoliation could occur not only in cells dominated by SBW host species (balsam fir and black spruce), but also in those dominated by other species, because preferred and less-preferred host species can be considered to be present as minor components in forest stands throughout the province (Bouchard and Auger 2014). This assumption was necessary due to data limitations and was intended to avoid underestimating the potential spread of outbreaks. Initial defoliation values were obtained from annual surveys assessing SBW defoliation intensity via low-altitude aerial photo interpretation (QMFFP 2022). Defoliation intensity maps were overlaid onto the study area grid, and each cell was classified as defoliated if at least 40% of its area overlapped with defoliated regions. Three defoliation intensity levels were employed: *low* defoliation corresponds to foliage loss in the upper third of some trees, *moderate* defoliation to foliage loss in the upper third of most trees, and *high* defoliation to foliage loss of the entire crown of most trees (QMFFP 2022). We defined four state variables related to SBW outbreaks for each cell: (1) current defoliation intensity ($defol_intensity_{x,t}$) in three levels (*low*, *moderate*, *high*); (2) cumulative defoliation ($cum_defol_intensity_{x,t}$) as the sum of defoliation intensities since the first year of consecutive defoliation, with intensity levels assigned as follows: *low* = 1, *moderate* = 2, and *high* = 3; (3) years of consecutive defoliation ($num_years_defol_{x,t}$); and (4) years without defoliation ($num_years_no_defol_{x,t}$), indicating outbreak cessation when five consecutive years lack defoliation (Gray 2008).

Environmental variables

Environmental variables used to model SBW outbreaks and forest dynamics included climate, soil texture, and elevation. Climatic data comprised annual precipitation ($Prec_{x,t}$) and annual mean temperature ($Tmean_{x,t}$). Since the model is projected to the future, future climate data (2021–2100) were provided through the PAVICS platform (Ouranos and CRIM 2018–2023) and were generated by the MIROC4 climate model following the CMIP5 protocol under the

Representative Concentration Pathway (RCP) 4.5 scenario. These climate data were obtained at the same spatial and temporal resolution as the SBW model. We selected this RCP scenario since it aligns with realistic climate projections and enables a balanced assessment of potential climate change impacts on SBW in the study area.

Soil texture ($soil_type_x$) was inferred from surficial deposits and drainage, which are critical drivers influencing regional forest dynamics (Saucier et al. 2009). These deposits fell into five primary categories: medium-textured (mainly glacial tills, > 25 cm deep), coarse-textured sandy deposits (mainly fluvioglacial, > 25 cm deep), fine-textured clay deposits (primarily glacio-lacustrine, > 25 cm deep), rock outcrops, or shallow deposits (depth < 25 cm), and organic deposits (containing > 40 cm of organic matter). Soil texture variable did not change with time. Data on surficial deposits, along with elevation data, were extracted from <https://www.donneesquebec.ca/> and used at the model cell resolution.

The model considers that SBW outbreak development depends on the local suitability of climate for SBW ($sbw_niche_{x,t}$). We determine temperature suitability classes by linking mean annual temperature with SBW defoliation data between 1967 and 1991 (QMFFP 2022), defining the temperature ranges most favorable for SBW activity. Based on this calibration, temperature suitability was classified as *optimal* ($0.5 \leq T^\circ \leq 2.8$), *good* ($-1.5 \leq T^\circ < 0.5$ or $2.8 < T^\circ \leq 4$), and *poor* otherwise. Using discrete classes facilitates the application of simple decision rules in simulations and avoids overinterpreting small differences in temperature values that are not supported by empirical evidence. These classes were also compared with results reported in previous studies looking at the SBW's climatic niche (Régnière et al. 2012; Boulanger et al. 2024).

The model also considers that the establishment of a dominant tree species during forest succession depends on local climatic and soil conditions. We defined a site quality variable ($site_quality_{x,t}$) derived from SDM developed for Quebec forests (Périé et al. 2014; Périé and de Blois 2016), which integrate annual precipitation, mean annual temperature, and soil texture to estimate species-specific habitat suitability. Suitability was categorized as high (25th–75th percentiles), moderate (5th–25th and 75th–95th percentiles), and unsuitable otherwise, according to

climatic envelopes. Full details on model calibration and validation are provided in the original studies.

Landscape processes

SBW outbreaks

Defoliation by SBW

The development of SBW outbreaks at the scale of the province of Quebec was simulated through four successive phases: pre-epidemic, epidemic, collapse and endemic (Chabot et al. 2014). In this study, we considered these phases to be synchronized at the scale of the province, reflecting the patterns observed during the twentieth century outbreaks (Blais 1983; Royama 1984; Navarro et al. 2018).

- The *pre-epidemic* phase, which lasts between 3 and 4 years uniformly across the landscape, corresponds to the emergence of few localized and often spatially clustered outbreak epicenters (Bouchard and Auger 2014; Chabot et al. 2014). At the scale of the province, these epicenters are more likely to emerge in areas that have not been recently defoliated, in highly suitable climatic conditions for the SBW, and at the bottom of valleys (Bouchard and Auger 2014; Chen et al. 2021). In this study, the *pre-epidemic* phase was represented by simulating the occurrence of epicenters in areas with a probability that was directly proportional to the number of consecutive years without defoliation (*num_years_no_defol*)—reflecting a greater likelihood of vegetation recovery from the previous outbreak and food accumulation for SBW (Blais 1983)—and inversely proportional to elevation, used as a proxy for valley bottoms. Epicenters were also more likely to emerge in locations where conditions were optimal for SBW (mean annual temperatures $0.5 \leq T^{\circ} \leq 2.8$). During each year of the pre-epidemic phase, we randomly selected between three and six epicenters using a uniform distribution. Each epicenter caused defoliation across 20 to 40 cells, within a broader area ranging from 200 to 2000 cells—corresponding approximately to a radius of 15 to 50 km (see Appendix S1).
- The defoliation intensity (*defol_intensity*) in cells affected for the first time and dominated by host species was randomly assigned as follows: *low* in 50% of the epicenter locations, *moderate* in 37%, and *high* in 13%. In affected cells dominated by non-host species, the defoliation intensity was considered *low*. These proportions were derived from historical defoliation maps, reflecting observed patterns in newly affected cells (see Appendix S1).
- In cells already affected during the previous year, defoliation intensity was derived from the following Eq 1

$$defol_intensity_{x,t} = host_pref_{x,t} * sbw_niche_{x,t} \quad (1)$$
- where *host_pref_{x,t}* represents the susceptibility of the host species within cell_{x,t} according to dominant tree species composition and age class, ranging from 0 to 1 (Appendix S1; Pothier et al. 2012; Bouchard and Auger 2014); and *sbw_niche_{x,t}* indicates the climatic suitability for SBW (Gray 2013) and can take the value 1, 0.5, or 0.2 when the SBW climatic niche is optimal, good, or poor, respectively, as exposed in the parameterization section and Appendix 1. Defoliation intensity was categorized as *low* when $0.1 < defol_intensity \leq 0.25$, *moderate* when $0.25 < defol_intensity \leq 0.50$, *high* when $defol_intensity > 0.50$, and otherwise the location was not defoliated that year (Appendix S1).
- The *epidemic* phase, during which SBW defoliation spreads from initial epicenters across the province (Chabot et al. 2014; Régnière et al. 2019a, b). The duration of this phase ranged between 9 and 12 years and was randomly selected using a uniform distribution. This phase identifies new cells at risk of being affected within a defined radius from cells defoliated during the previous year. The outbreak radius was sampled from a truncated exponential-like distribution biased toward smaller values, bounded between 25 and 35 km. The range of potential radii was determined from historical SBW defoliation data and based on Gandiaga and James 2023, as detailed in Appendix S1. *Pspread_{x,y}* determines the probability of successful spread from a source cell (x) defoliated during the previous year to a target cell (y), and is calculated as Eq. 2

$$Pspread_{x,y} = wSpp * PSpp_x + wDist * (1 - Dist_{x,y}) + wWind * (Angle_{x,y} * WindDir) \quad (2)$$

- where $PSpp_x$ is a parameter associated with the dominant species in the source cell x , set at 0.3 for black spruce and 0.7 for balsam fir. $Dist_{x,y}$ represents the distance between the source cell and the target cell, rescaled between 0 and 1 (0 corresponds to adjacent cells, while 1 corresponds to cells at the maximum distance within the defined radius). $Angle_{x,y}$ indicates the relative angle between the source and target cells. Finally, $WindDir$ indicates the wind direction, giving a probability [0, 1] of SBW to arrive at the target cell considering the wind effect. In the study area, the prevailing wind directions are primarily from west to east, increasing the likelihood of SBW reaching cells located to the east of the source cells. Weights for each factor ($wSpp$, $wDist$, and $wWind$) are set to 0.15, 0.15, and 0.7 (Appendix S1), respectively, reflecting the influence of each factor.
- For newly defoliated cells dominated by host species, defoliation intensity was assigned as in the *pre-epidemic* phase. For non-host species, defoliation was randomly assigned as *low* in 85% and *moderate* in the 15% of the newly affected cells (Appendix S1). Defoliation intensity of cells disturbed during previous years was defined by Eq 1.
- The *collapse* phase, during which SBW populations begin to decline synchronously across the province, lasts between 3 and 6 years (Chabot et al. 2014), with this duration selected from a uniform random distribution. During this phase, only a small number of newly defoliated cells (between 10 and 20) are randomly selected and assigned *low* or *moderate* defoliation intensities (Appendix S1). The outbreak decrease until the number of defoliated cells across the province reaches zero. To achieve this, we set the defoliation intensity to zero in cells that have experienced 18 consecutive years of defoliation ($num_years_defol_{x,t} \geq 18$), under the assumption that SBW resources would be depleted at this point (Chabot et al. 2014).
- The *endemic* phase represents a period when SBW population levels are very low, with sporadic and minor defoliation on trees. This phase lasts approximately 15 years on average (Blais 1983; Royama 1984). During simulations, this dura-

tion was drawn from a Gaussian distribution with mean = 15 years and standard deviation = 2 years, and applied across the whole province (Appendix S1).

SBW-caused tree mortality

Stand mortality caused by SBW was assumed to occur only during the epidemic phase. Each year, the model identifies a set of vulnerable cells based on the dominant species (balsam fir and black spruce) and the number of consecutive years of defoliation. For balsam fir-dominated cells, 25% of the cells that had experienced 5–7 consecutive years of defoliation, 40% of the ones with 8–10 consecutive years, and 70% of the ones with more than 10 consecutive years were selected randomly. For black spruce dominated cells, 5% of the cells with 8–10 years of consecutive defoliation, and 15% for those with more than 10 years were also selected randomly (based on Pothier 2012, see Appendix S1). The probability of mortality among these selected cells, $Pmortality_{x,t}$, was calculated as follows Eq. 3

$$Pmortality_{x,t} = cum_defol_intensity_{x,t} * defol_intensity_{x,t} \quad (3)$$

Where $cum_defol_intensity_{x,t}$ is the sum of annual intensities since the first year of consecutive defoliation and $defol_intensity_{x,t}$ is the intensity of defoliation during year t . The values of $Pmortality$ were scaled between 0 and 1 for balsam fir and black spruce cells separately, considering the minimum and maximum probabilities across all evaluated cells. When a cell experienced mortality, stand age was set to 0 and the new dominant species was determined according to vegetation succession rules (Vegetation succession).

Harvesting simulation

Forest harvesting is an important driver of forest change in the study area. In Quebec, around 2000 km² of forests are harvested annually, corresponding to less than 1% of the total forested area (QMFFP, 2018). Clearcutting (< 10% tree retention at the stand level) is the primary mode of harvest in coniferous forests dominated by species like black

spruce and balsam fir, as well as in hardwood forests dominated by shade intolerant species such as aspen, with rotations that vary between 70 and 100 years. Although partial cuts (50–70% retention at the stand level) are also applied in Quebec, particularly in stands dominated by tolerant hardwoods, only clearcutting was considered as a harvesting strategy in this study.

Harvesting rates were defined for each Forest management unit (FMU). The annual clear-cut area was set as a fixed percentage (according to the tested scenarios, Climatic and management scenarios) of the managed forested area within each FMU and was applied only in mature stands. Stand maturity ranged between 60 and 120 years, depending on the dominant species. We assumed that clearcutting could occur in all forested area outside protected areas, except in cells dominated by shade tolerant species such as sugar maple.

Plantation-based regeneration

Currently, approximately from one-half to two-thirds of the clear-cut areas in Quebec are left to regenerate naturally, while the remaining are replanted (QMFFP 2018). In this study, we examined how the targeted regeneration of host or non-host species through plantation or other means could change outbreak patterns. We included scenarios with targeted regeneration of 50% of the harvested cells, by regenerating either black spruce or trembling aspen in cells that are suitable according to climate and soil type (*site_quality*). Black spruce is a host of the SBW but tends to tolerate defoliation relatively (Fuentelba et al. 2019), and regenerating this species can be considered an appropriate strategy for mitigating SBW-caused mortality while also retaining a commercially interesting species for the wood industry (Sainte-Marie et al. 2015). Currently, this species is widely used in reforestation programs on public lands, particularly in the boreal zone. Trembling aspen is not typically employed in plantations in Quebec. Nevertheless, we tested the potential of promoting the regeneration of this deciduous non-host species for reducing outbreak intensity and spread.

Vegetation succession

Vegetation succession in our model encompassed two types of natural forest transitions: (1) transitions to late-seral compositions occurred when the stand age exceeded the maturity age by at least 40 years; and (2) post-disturbance succession after cell mortality resulting from the impact of the SBW or harvesting. In both cases, the selection of new species in a given cell was based on transition probabilities that varies according to the specific processes involved (aging, SBW, or harvesting). These probabilities were derived from studies conducted across the study area (Saucier et al. 2009) and expert judgments (Bouchard et al. 2019; Appendix S2 Tables). The selection of new species also considered factors such as climate and soil suitability (*site_quality*), species composition in the surrounding cells, and the maximum colonization distances of each species, as defined in Bouchard et al. (2019).

Model parameterization and validation

Model input parameters—including the number and size of SBW outbreak epicenters, spread radius, and defoliation intensities across SBW outbreak phases—were calibrated using SBW defoliation data from 1968 to 2020 (QMFFP 2022). To estimate the number and size of annual epicenters, we performed a spatial analysis of historical defoliation maps, grouping adjacent defoliated polygons within 80 km and calculating centroid-to-boundary distances (both minimum and maximum) for each epicenter (Appendix S1; Figure S1.1). Defoliation intensities of newly defoliated cells were derived by intersecting historical SBW maps with forest inventory data, allowing us to determine the relative frequency of low, moderate, and high defoliation across host and non-host species, and to differentiate between newly or previously defoliated cells, as well as between pre-epidemic and epidemic phases (Appendix S1; Figure S1.2). Defoliation intensities in previously affected cells were estimated from host susceptibility (based on data published on Pothier et al. 2012) and SBW niche parameters, with thresholds defining defoliation categories. We evaluated these parameters and thresholds in sensitivity tests by projecting 2021 defoliation and comparing predictions with observed data (Appendix S1, Figure S1.4). Spread radius was estimated by

computing the distances between newly defoliated patches and those from the previous year (Appendix S1; Figure S1.1). Outbreak durations and phase lengths were based on historical records and literature (Royama et al. 1984; Chabot et al. 2014; Jardon et al. 2003; Boulanger and Arseneault 2004). Probability of mortality was adapted from Pothier et al. (2012).

To capture the role of host preference, distance, and wind direction in SBW spreading (Eq. 2), we performed a sensitivity analysis of the spread weights (w_{Spp} , w_{Dist} , and w_{Wind}), testing alternative parameter combinations and comparing the resulting spread patterns in terms of elongation and mean epicenter polygon size with historical SBW defoliation maps (2004–2020, QMFFP 2022; Appendix S1; Figure S1.3).

To validate our model, we simulated the initial outbreak development phases (pre-epidemic and epidemic) of the current SBW outbreak in Quebec (2006–2020) and compared the outputs to observed defoliation data from the same period (QMFFP, 2022). Ten stochastic runs were performed, considering no forest management and current climatic condition. For each year and defoliation intensity level (*low*, *moderate*, *high*), we compared the simulated and observed defoliated areas, assessing temporal agreement using Spearman's correlation and root mean square error (RMSE). Results are shown in Appendix S3. This validation provides confidence in the model capacity to reproduce outbreak dynamics under long-term scenario simulations.

Climatic and management scenarios

We defined different scenarios to examine the combined effects of varying climate, harvest rates and

reforestation strategies on future outbreak patterns as shown in Table 1. In this model, climate influences SBW dynamics by altering landscape suitability (sbw_niche), which affects the emergence of outbreak epicenters and defoliation intensity. It also modifies suitability for the dominant tree species ($site_quality$), thereby influencing natural succession probabilities. Two climate scenarios were considered: absence (Stable Climate, SC) and presence of climate change using the RCP 4.5 scenario (CC; Table 1).

We also tested how varying forest harvest and regeneration parameters affect SBW dynamics and impacts. These include a range of annual harvest rates (from 0 to 1% of the total forest area per year), with 1% being a realistic rate for the province (QMFFP 2018). In addition, we examined the effect of post-harvest regeneration assumptions, comparing scenarios that favor the establishment of non-host species such as trembling aspen (80% of plantations), with more conventional scenarios in which harvested areas are dominated by the plantation of a secondary host species, black spruce (80% of plantations).

Although some of the tested scenarios may not reflect current forest management practices in Quebec, they were designed to span a plausible range of conditions and parameter values. Exploring such contrasted yet ecologically realistic scenarios is essential to identify thresholds or nonlinear responses in SBW dynamics, and to assess the potential leverage of forest management under varying climate conditions.

Analysis

Simulations were conducted over an 80-year period from 2020 to 2100 with a one-year time step. Each scenario was repeated 10 times to account for

Table 1 Definition of scenario parameters in terms of climate change, harvesting rate, planting rate, and proportion of planted species

Planting	Harvesting rate			Climate change
	0%	0.50%	1%	
none	SC_H0	–	–	No
	CC_H0	–	CC_H1_P0	Yes
50% (20% BS–80% TA)	–	CC_H0.5_P50_TA	CC_H1_P50_TA	
50% (80% BS–20% TA)	–	CC_H0.5_P50_BS	CC_H1_P50_BS	

Codification: SC stable climate, CC climate change, H harvesting ratio, P planting ratio, BS black spruce, TA trembling aspen. The table shows the code name of the seven defined scenarios

stochasticity in the simulated processes, including the duration of SBW outbreak phases, the random selection of outbreak epicenters in terms of number and spatial extent, the probabilistic spread of defoliation across the landscape, the assignment of defoliation intensity levels, the selection of cells induced to die, the selection of cells targeted for management, and the selection of post-disturbance successional species. This number of repetitions was chosen as a compromise between computational feasibility and capturing the variability among simulations, as well as we kept consistency with previous studies using similar models (e.g., Bouchard et al. 2019). At each iteration, we recorded the following model output variables to assess SBW impact on the forest landscape: annual defoliated area (Dy) and SBW-induced mortality area (My) in km^2 , and their respective total accumulated values over the simulation period (Dp and Mp). Additional output variables included species composition (area dominated by each species in km^2), northern latitude of defoliated and mortality-affected cells (in meters, using EPSG: 2154 projection), and SBW defoliation intensity (categorized as *low*, *moderate*, or *high*).

To evaluate the effects of harvesting levels and plantation-based regeneration strategies, we applied a Repeated Measures ANOVA with annual defoliated area (Dy) and mortality area (My) as response variables, using year and scenario as fixed factors, while simulation run was included as a random factor. This allows modeling the correlation structure of repeated measures from the same simulation run over time (Bates et al. 2015). Multiple comparisons between scenarios were performed using a post-hoc Tukey test. Differences in SBW defoliation intensity across scenarios were assessed using a Chi-square test, comparing average defoliation values per scenario across defoliation intensity. To examine the impact of harvesting rates on SBW-induced defoliation and mortality (Dp and Mp), we fitted a linear mixed-effects model that included harvest rate as a continuous numerical predictor, testing both linear and quadratic effects to account for potential non-linearities, and scenario as fixed factor.

Finally, we analyzed annual defoliated area (Dy) by latitude to investigate climate change effects on the spatial distribution of SBW outbreaks. For this, we used a Mixed-Effects Model with the latitude where SBW outbreaks epicenters were located (in meters) as

the response variable, year and scenario as fixed factors, and simulation run as a random factor. Analyses were conducted using the *aov* and *lmer* functions from the *stats* and *lmerTest* packages in R (R Core Team), respectively, and we set the significance level at $\alpha=0.05$.

Results

Our model simulated SBW dynamics across the province of Quebec over an 80-year period and a 589,856 km^2 landscape. Simulations predicted three SBW outbreaks and associated tree mortality from 2020 to 2100 across all scenarios (Fig. 3). The first episode, already underway at the start of the simulations, occurred in the 2020s. The second episode took place in the 2050s, followed by a third in the late 2080s to early 2090s. The onset and duration of each episode varied across scenarios and stochastic simulation runs. The maximum annual extent of defoliated area ranged from 60,000 to 160,000 km^2 (11–31% of forested area), while SBW-induced mortality affected between 3000 and 9000 km^2 (0.8–2.8% of host-dominated forest area, the only cells where mortality is allowed).

The validation process confirmed that the model successfully captured the magnitude and temporal trends of the current outbreak (2006–present; Appendix S3.). The Spearman correlation between observed and simulated annual defoliated area was high ($\rho=0.98$), with a root mean squared error (RMSE) of 6,020 km^2 . When defoliation intensity was considered separately (*low*, *moderate*, *high*), correlations ranged from 0.82 (high intensity) to 0.98 (moderate intensity), and RMSE values ranged from 15,400 km^2 (low intensity) to 2300 km^2 (moderate intensity), confirming the model's reliability across intensity levels.

Simulation outputs also reflected that tree species composition shifted over time according to both climate and management scenarios (Fig. 4). Black spruce, initially the most abundant species, showed a gradual decline throughout the simulation period (showing a maximal reduction of 45%). This pattern was especially evident in scenarios with higher harvesting intensity, where the decline in black spruce was accompanied by lower SBW-induced mortality (Effectiveness of forest harvesting and plantation-based regeneration on SBW). In contrast, scenarios

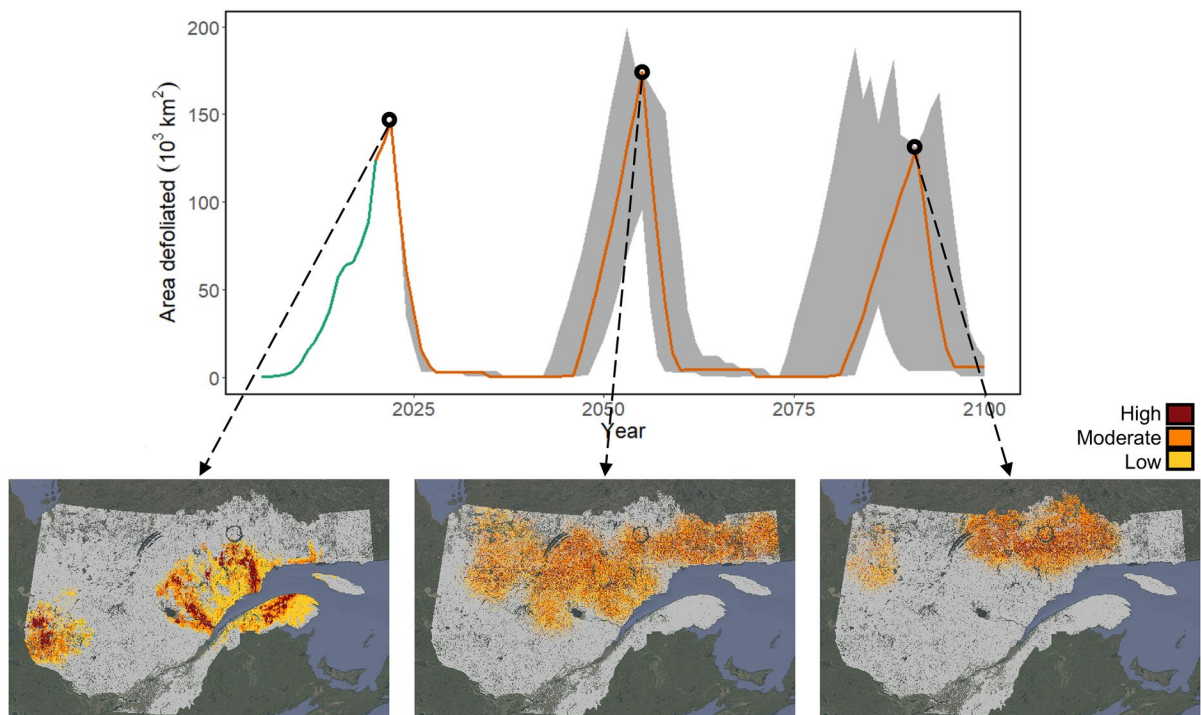


Fig. 3 Defoliation dynamics and distribution. Defoliated area by SBW from 2020 to 2100. The curve displays historic data in green and one simulation run of projected data under CC_H0 scenario in orange. The gray envelope represents the range of variations obtained for 10 simulation runs of the same

scenario. The maps show the distribution of SBW defoliation when it reaches its maximal extent for each of the three outbreaks (2020, 2051, and 2084) in one given simulation run. The colors indicate the defoliation intensity level

with minimal harvesting showed higher host abundance and larger areas affected by SBW defoliation and mortality. The area occupied by balsam fir was higher in scenarios without harvesting (SC_H0 and CC_H0), or with harvesting but no planting (CC_Hn_A0) (Fig. 4). Aspen area increased by the end of the simulations in managed scenarios, with a more substantial increase at higher harvesting rates, reaching up to five times its initial area (Fig. 4).

Influence of climate change

In the no climate change scenario (SC_H0), the mixed-effects model showed that the latitude of SBW defoliation events remained relatively constant throughout the century (Estimate=162.3, $t=30.9$, $p=0.98$). Under climate change scenarios, however, there was a significant northward shift in defoliation episodes over time (Estimate=2558, $t=481.6$, $p<0.001$; Fig. 5.A). This shift extended

beyond the northern limit of the managed forest area, where no management interventions were applied. The proportion of defoliated cells located beyond this northern limit rose from 1% of all defoliated cells in the 2020s peak to 27% in the 2050s peak, and 38% in the 2080s peak under the CC_H0 scenario (Fig. 5.B).

Defoliation and mortality due to SBW were significantly higher under the climate warming scenario compared to the stable climate scenario ($F_{1,1394}=7.24$, $p=0.007$ for defoliation and $F_{1,1394}=5.52$, $p=0.019$ for mortality; Fig. 5). In both the SC_H0 and CC_H0 scenarios, the 2020s peaks in defoliation and mortality were similar in magnitude. However, during peak outbreak periods in the 2050s and 2080s, the extent of defoliation and mortality were 15–50% higher under the CC_H0 scenario compared to conditions without climate change (Fig. 6).

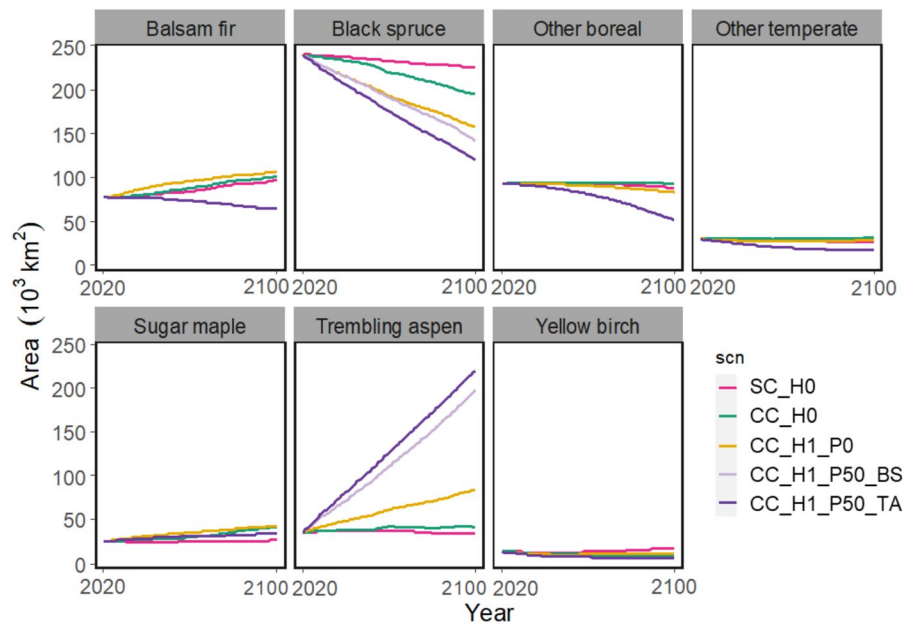


Fig. 4 Dominant species dynamics. Total area occupied by each dominant species throughout the 80 years of simulation. The mean of 10 runs for each scenario is represented with different colors. Only those scenarios more representative of species dynamics are plotted: SC_H0: baseline scenario without climate change; CC_H0: climate change without harvesting;

CC_H1_P0: climate change with harvesting but no planting; CC_H1_P50_BS: climate change with 1% of harvesting and 50% of planting, mainly black spruce; CC_H1_P50_TA: climate change with 1% of harvesting and 50% of planting, mainly trembling aspen

Effectiveness of forest harvesting and plantation-based regeneration on SBW

The different forest management scenarios (i.e., timber harvesting levels and planted species) did not significantly influence defoliated area (Figs. 6 and 7). Yearly defoliated area (D_y) showed no significant difference among scenarios overall ($F_{5,4186}$, $p = 0.402$) or in pairwise comparisons (Table 2), including comparisons between scenarios with the same harvest level but different plantation-based regeneration strategies. However, as expected given the model design, year had a consistently significant effect ($F_{80, 4186}$; $p < 0.05$), indicating temporal variation in defoliation patterns across scenarios. When differentiating the defoliation intensity levels (*low*, *moderate*, *high*) as categorical response variables results also showed no significant differences across scenarios ($X^2_{10} = 15.4$; $p = 0.12$).

In contrast, the management scenarios had a significant effect on SBW-induced forest mortality. Comparing unmanaged scenario (CC_H0) to scenarios with moderate or high harvesting combined with

plantation-based regeneration favoring trembling aspen (CC_H0.5_P50_TA and CC_H1_P50_TA), yearly mortality area (M_y) was significantly higher in the absence of management (Table 2). Peaks in mortality in the 2020s and 2050s were similar across scenarios, but by the 2080s, peak mortality was 30–40% higher in the no-management CC_H0 scenario compared to the CC_H0.5_P50_TA and CC_H1_P50_TA scenarios. No significant differences were observed in M_y between scenarios prioritizing trembling aspen plantation and those prioritizing black spruce, though black spruce scenarios tended to show slightly higher defoliation and mortality levels (Table 2, Fig. 5). Year was also a significant factor in all pairwise mortality comparisons ($F_{80, 4186}$; $p < 0.05$).

Cumulative defoliation and mortality over the simulation period (D_p and M_p) both declined progressively as harvesting levels increased across scenarios (Fig. 7). A linear mixed effect analysis revealed a non-significant negative effect of harvest level on defoliation (Estimate = $-9.45e+04$, $t = -1.08$, $p = 0.279$), while a significant negative effect was found for mortality due to SBW (Estimate = -8098 ,

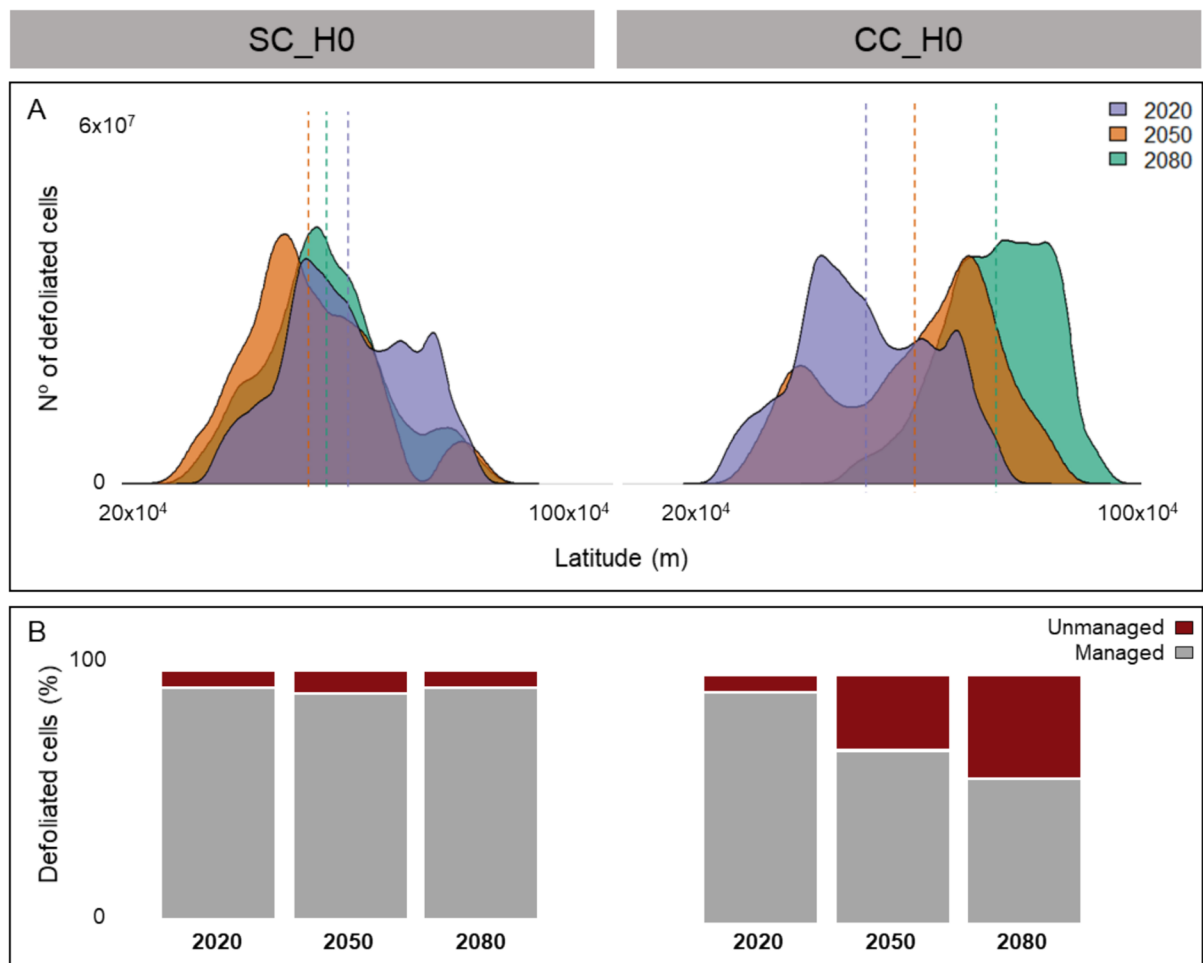


Fig. 5 Defoliation across latitude. (A) Mean number of cells defoliated by SBW across 10 simulation runs, shown by latitude (in meters, EPSG: 2154 projection) for three SBW outbreaks over the century, under the SC_HO scenario (left) and CC_HO_A0 scenario (right). Dashed lines indicate the mean

latitude of each outbreak across all runs. (B) Mean percentage of defoliated cells located within the managed or northern unmanaged forest zones were averaged over the 10 simulation runs, under both climate scenarios

$t = -2.181$, $p = 0.03$). The quadratic term, however, was not significant in any case.

Discussion

Understanding the interplay between forestry practices and the dynamics of major forest insects, such as the spruce budworm in Eastern North America, is crucial for designing effective and sustainable forest management practices. It is especially relevant in regions where forestry is both an economic pillar and a landscape-scale change driver. Our study used

a spatially explicit model of SBW dynamics to assess how different management strategies—varying in harvesting levels and planting of secondary and non-host species—could impact forest defoliation and tree mortality across a large region.

Climate change effects on spruce budworm-caused mortality.

The magnitude of projected SBW impacts in Quebec is substantial. In our simulations, peak annual defoliated areas ranged from 60,000 to 160,000 km² (11–31% of all forested area), comparable to the

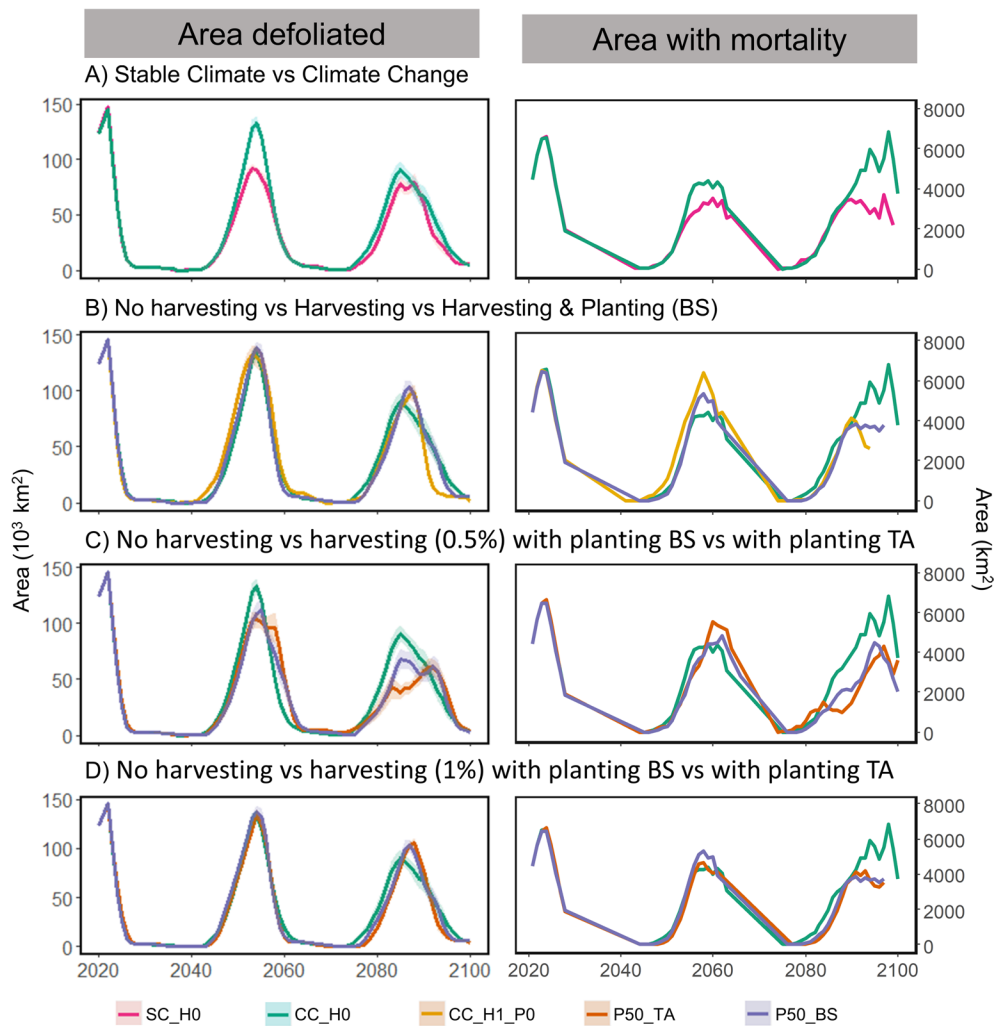


Fig. 6 Projected defoliation and induced mortality dynamics. Total forest area affected by defoliation (left columns) and mortality (right columns) under different management and climate scenarios. (A) Comparison between the baseline scenario (SC_H0) and the climate change scenario (CC_H0) without harvesting. (B) Effects of different management approaches under climate change: no management (CC_H0), harvest-

ing without planting (CC_H1_P0), and harvesting with black spruce planting (CC_H1_P50_BS). C, D) Scenarios with harvesting rates of 0.5% and 1%, respectively, under planting primarily black spruce (CC_Hn_P50_BS) or planting primarily trembling aspen (CC_Hn_P50_TA), compared to the no-management scenario (CC_H0)

cumulative defoliation of approximately 120,000 km² observed during the ongoing 2006–2022 outbreak in Quebec (QMFFP 2022). SBW-induced mortality affected 3,000–9,000 km² per peak episode (0.8–2.8% of host forests), which is consistent with the scale of damage projected in other regional modeling studies (e.g., Gray and Re 2000). Alongside wildfire—which, for example, burned approximately 45,000 km² in 2023 in Quebec (Boulanger 2025)—SBW outbreaks remain a dominant driver

of forest dynamics and structural change across the whole territory.

Our study projects a potential northward expansion of SBW outbreaks driven by climate change, consistent with past shifts in outbreak latitude documented throughout the nineteenth and twentieth centuries (Régnière et al. 2012; Cooke 2024), temperature being a key driver of SBW development (Pureswaran et al. 2015; Boulanger et al. 2024). In Quebec, the northern range of SBW defoliation is probably

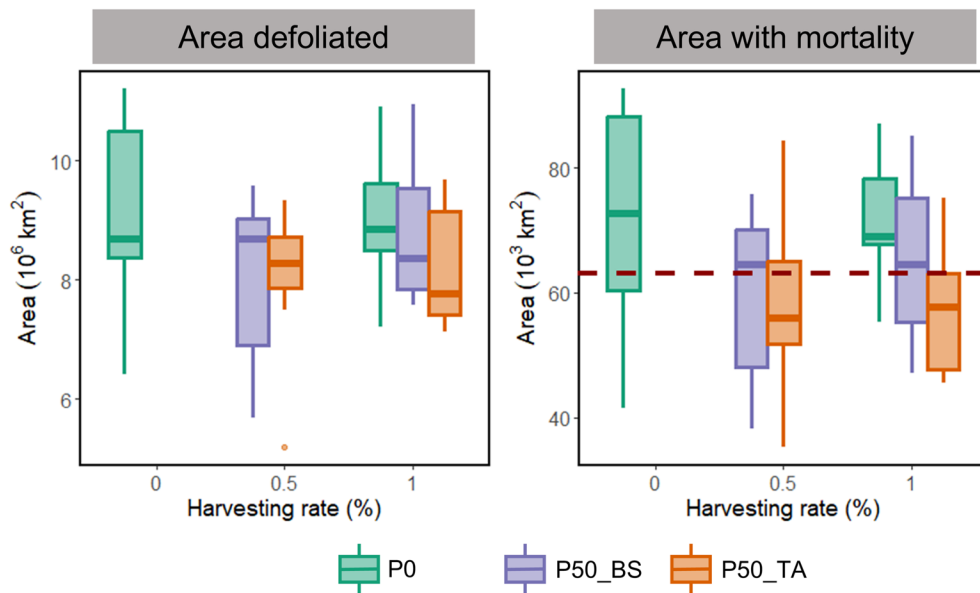


Fig. 7 Accumulated defoliation and induced mortality. Box plots showing the total accumulation of forest area defoliated (Dp ; left panel) and forest area experiencing mortality (Mp ; right panel) over an 80-year simulation period based on each management scenario. Scenarios with no planting are repre-

sented in green; scenarios favoring black spruce in plantation-based regeneration, in purple; and the scenarios favoring trembling aspen, in orange. The red dashed line indicates the threshold beyond which the scenarios are significantly different from the no management scenario (CC_H0)

currently constrained by unfavorable climatic conditions more than by a lack of host species (Gray 2008; Candau and Fleming 2011; Boulanger et al. 2024). Our results suggest that climate change will facilitate SBW spread in northern-forested regions where balsam fir, its primary host, is less dominant, but black spruce—a secondary host—prevails. As shown in this study, a northward expansion of fir distribution and abundance is likely to take place eventually, but will probably occur at a much slower pace than SBW outbreak northward progression (Boulanger et al. 2024).

This northward shift will increasingly affect unmanaged forests (Pureswaran et al. 2015). In our projections, the proportion of defoliated cells located to the north of the managed forest zone increased sharply from only 1% during the 2020 peak to 38% by the 2080 peak. This trend suggests that the geographic focus of management strategies may need to shift over time; reliance on historical southern management alone will likely be insufficient. Developing adaptive strategies for northern regions could become relevant, but implementing silvicultural interventions in these remote areas poses ecological and logistical challenges and would require a rethinking of timber

supply zones and infrastructure planning. Although black spruce has a lower mortality risk from SBW, its prevalence in newly impacted areas implies that mortality levels could remain relatively low over time. Climate change may narrow the phenological mismatch between SBW and black spruce (i.e., larval development and tree buds opening) and increase mortality for this species (Pureswaran et al. 2015), but this potential effect was not examined in this study.

Effect of forest management on outbreak patterns

While forest management practices did not significantly alter the total defoliated area due to SBW, they substantially affected forest mortality patterns. This is likely due to the fact that defoliation can occur in any cell containing host species, regardless of their relative abundance, whereas mortality is restricted to cells where host species represent a dominant component of the stand. Interestingly, this pattern suggests a potential trade-off between harvesting intensity and SBW-induced mortality: in scenarios with minimal harvesting, the increased abundance of host species may support larger SBW populations and

Table 2 Pairwise comparisons of scenarios from post-hoc Tukey tests following a repeated measures ANOVA on annual defoliated area and mortality area under different harvesting intensities and planting strategies

Scenario	Variable	Scenario			CC_H1_P50_BS			CC_H1_P50_TA			CC_H0.5_P50_BS			CC_H0.5_P50_TA		
		Mean. Diff	95% CI	p-val	Mean. Diff	95% CI	p-val	Mean. Diff	95% CI	p-val	Mean. Diff	95% CI	p-val	Mean. diff	95% CI	p-val
CC_HO	Def	88.95	(- 1609.02; 1786.93)	0.887	- 223.91	(- 1921.89; 1474.06)	0.825	- 741.55	(- 2439.52; 956.43)	0.273	- 940.17	(- 2638.15; 757.80)	0.152	- 900.86	(- 2598.83; 797.11)	0.174
	Mort	3.47	(- 257.21; 264.165)	0.972	- 89.51	(- 350.19; 171.18)	0.367	- 198.01	(- 458.70; 62.67)	0.032*	- 162.90	(- 423.53; 97.79)	0.084	- 189.37	(- 450.05; 71.32)	0.043*
CC_H0.5_P50_TA	Def	- 811.91	(- 2509.88; 886.07)	0.135	676.95	(- 1021.03; 2374.92)	0.252	159.31	(- 1538.66; 1857.29)	0.781	- 39.31	(- 1737.29; 1658.68)	0.946			
	Mort	- 185.89	(- 446.58; 74.79)	0.044*	99.86	(- 160.83; 360.54)	0.261	- 8.65	(- 269.33; 252.04)	0.917	26.47	(- 234.22; 287.15)	0.762			
CC_H0.5_P50_BS	Def	- 851.22	(- 2549.19; 846.75)	0.117	716.26	(- 981.71; 2414.23)	0.223	198.63	(- 1499.35; 1896.60)	0.728						
	Mort	- 159.42	(- 420.11; 101.26)	0.084	73.39	(- 187.30; 334.07)	0.41	- 35.11	(- 295.80; 225.57)	0.677						
CC_H1_P50_TA	Def	- 652.59	(- 2350.57; 1045.38)	0.218	517.63	(- 1180.34; 2215.61)	0.38									
	Mort	- 194.54	(- 455.22; 66.15)	0.033*	108.50	(- 152.18; 369.19)	0.217									
CC_H1_P50_BS	Def	- 134.96	(- 1832.93; 1563.07)	0.717												
	Mort	- 189.37	(- 450.05; 71.32)	0.359												

The table reports mean differences, 95% confidence intervals, and *p*-values. Statistically significant differences ($p < 0.05$) are indicated in bold. *Def* refers to defoliated area, *Mort*. refers to mortality area

result in higher stand-level mortality. Conversely, more intensive harvesting reduces host availability and may therefore constrain the extent of mortality, even if defoliation remains similar. These effects are particularly evident in scenarios involving no management, where an increase in the abundance of late-successional balsam fir cells is observed, which leads to higher SBW-caused stand mortality. In contrast, scenarios incorporating the plantation of a non-host, trembling aspen, SBW-induced stand-level mortality was lower. Defoliation may still cause tree-level mortality in stands dominated by non-hosts (Bouchard et al. 2007; Sainte-Marie et al. 2015), but this was not examined in this study. On the other hand, prioritizing the plantation of black spruce (scenarios CC_H0.5_P50_BS and CC_H1_P50_BS) did not significantly reduce tree-induced mortality.

Beyond their direct effects on host availability, management practices also influence the spatial configuration of the landscape, which can alter outbreak dynamics. By changing the distribution and connectivity of host patches, harvesting and regeneration decisions affect how defoliation spreads across the forest matrix (Robert et al. 2018), reinforcing the importance of management as a large-extent driver of disturbance regimes.

Finally, harvesting combined with the plantation of non-hosts delayed the most severe SBW impacts in the latter half of the century, particularly around the projected outbreak peak in the 2080s. This lag reflects the time required for management interventions to accumulate and produce noticeable shifts in forest composition at the landscape scale.

Implications for plantation-based forest management

Forest management faces the dual challenge of reducing insect-induced impacts while maintaining long-term ecosystem services. Our simulations provide quantitative evidence of how management can mitigate the intensification of SBW-induced mortality. We also observed that climate change will shift SBW activity northward, in regions that are largely unmanaged, leading to reduced co-occurrence with its primary host, balsam fir, but increased overlap with its secondary host, black spruce that is more tolerant to defoliation. This raises important questions about the potential use of forests management to mitigate SBW impacts: while silvicultural investments to mitigate

SBW impacts in the current distribution of the insect might prove unnecessary, investments in the regions where the insect is likely to have more impacts in the future might not be possible considering constraints to forest management (Boulanger 2019).

However, forest management aimed at controlling insect outbreaks might imply other risks and faces the challenge of maintaining the provision of key forest ecosystem services (Miller and Rusnock 1993). In the context of historic SBW outbreaks, the silvicultural hypothesis suggests that historical logging activities, often driven by economic criteria, have altered natural successional trajectories, promoting landscapes dominated by balsam fir and thereby increasing susceptibility to SBW outbreaks (Robert et al. 2012). To limit SBW impacts, management strategies should therefore promote adjustments in species composition, age structure, stand density, and the selective removal of the most vulnerable stands (Cappuccino et al. 1998; Muzika and Liebhold 2000; Pureswaran et al. 2016). However, while removing preferred host species may effectively reduce susceptibility, it can also have short-term ecological trade-offs and uncertain long-term outcomes (Spittlehouse and Stewart 2004; Malmshamer et al. 2008). Our results show that reforestation policies promoting non-host species like trembling aspen can reduce SBW-induced forest mortality, but their effectiveness depends on long-term forest development trajectories and the broader climatic context.

Overall, our results highlight that forest management can mitigate SBW-induced mortality without altering defoliation patterns. This suggests that strategic interventions—especially those modifying species composition—can reduce the ecological and economic impacts of SBW outbreaks. Given the projected northward expansion of SBW under climate change, such strategies become increasingly relevant in remote boreal regions, where implementation is challenged by accessibility and infrastructure. By integrating climate, forest dynamics, and pest spread, our modeling framework supports the development of regionally adapted strategies that align pest control with broader forest sustainability goals. These insights underscore the importance of explicitly incorporating insect dynamics into long-term forest management planning and policy design. While other processes such as fire, wind throw, or outbreaks of different insects could alter the magnitude and spatial

patterns of impacts, they are unlikely to overturn the general conclusion that decreasing host availability reduces SBW vulnerability at the landscape scale.

Model limitations and potential

The model presented in this study simulates the dynamics of SBW outbreaks, forest management, and allows projections under climate change conditions. Nonetheless, several limitations in its current implementation must be acknowledged. The model focuses exclusively on SBW and does not include other major natural disturbance agents such as wildfire, which strongly shape forest composition and structure in Quebec (Bouchard et al. 2019; 2023). Fire typically favors black spruce and hardwood expansion while reducing balsam fir abundance, which would likely mitigate SBW impacts in certain areas (Bergeron and Leduc 1998). Integrating a fire disturbance module would therefore improve the realism of landscape-scale dynamics by capturing both direct effects and interactions with insect outbreaks and management (Sturtevant et al. 2004; 2012; Bouchard and Pothier 2008; James et al. 2011; 2017).

Additionally, model parameterization involves simplifications and data limitations. For example, in the model, SBW spread between cells is influenced by factors including tree species composition, the distance between cells, and wind direction. Although the current parametrization assumes a predominant westward wind direction across the entire study area, spatially varying wind data could refine dispersal estimates. Another constraint resulted from the fact that only the dominant species and stand-level mortality were examined in this study. Incorporating finer stand-level information, including not only the main forest species but also secondary ones and relative or absolute abundances, would provide a more comprehensive understanding of how species composition influences SBW population dynamics (James et al. 2011), and potentially lead to more accurate evaluations of SBW effects on harvestable volumes or carbon dynamics.

Although model validation showed good agreement with observed SBW dynamics—yielding high Spearman correlations and relatively low RMSE values—some limitations remain. Validation was based on an outbreak that is still ongoing, meaning only the initial and part of the epidemic phases are currently

observable. Notably, a collapse in high-intensity defoliated areas after 2020 in parts of the study area (QMFFP 2022) was not consistently reproduced in simulations, suggesting that further refinement is needed to better capture more local dynamics instead of assuming province-level synchrony (Cooke 2024). Additionally, defoliation intensity was unaffected by the management treatments applied in scenarios, indicating that the model may not fully capture the potential of shaping SBW intensity at the landscape scale. Improving this aspect will require both a better understanding of outbreak drivers and finer input data—such as the presence of host species among secondary forest components within each cell.

Despite these limitations, the model can serve as a valuable tool to test specific management strategies by separating their effects on specific phases of outbreak development, taking into account both current conditions and projected climatic scenarios. These strategies could focus on age structure management or aim to disrupt the connectivity of SBW-susceptible stands, thereby reducing their colonization capacity (Robert et al. 2018), something that was not explored in this study. Furthermore, this modeling approach—separating outbreak development into distinct phases—could also be applied to other insect species exhibiting similar cyclic dynamics. The model's ability to integrate interactions between climate, forest composition, and disturbance regimes makes it a useful framework for exploring how different pests might spread and respond to changing environmental conditions.

Conclusions

Our results demonstrate that while forest defoliation by SBW is not significantly influenced by forestry practices, forest SBW induced mortality can be reduced by accurate management practices, especially those that modify forest composition through harvesting and subsequent planting of non-host species like hardwoods. Projected climatic changes over the next 80 years indicate a continued northward shift in SBW distribution, leading to reduced co-occurrence with its primary host, balsam fir, but increased overlap with its secondary host, black spruce. This raises important questions about the

potential use of forests management to mitigate SBW impacts: while silvicultural investments to mitigate SBW impacts in the current distribution of the insect might prove unnecessary, investments in the regions where the insect is likely to have more impacts in the future might not be possible considering constraints to forest management. The model presented in this study enables the investigation of the primary landscape drivers in Quebec, allows for the incorporation of additional factors such as fire, and serves as a foundation for testing specific insect management strategies.

Acknowledgements This research was funded through the FRQNT Team Research Project 286905 to PJ, EF, MB. QC acknowledges the support from the Juan de la Cierva Program (Spanish Ministry of Science and Innovation – Juan de la Cierva, 2021).

Author contributions Q.C. contributed to the conception, design, code elaboration, analysis, and drafting of the manuscript. N.A. participated in the conception, design, data collection, code elaboration, and revision of the manuscript. E.F. and M. B. collaborated on the conception, design, code elaboration, and revision of the manuscript. P.J. contributed on the revision of the manuscript.

Funding Open Access funding provided thanks to the CRUE-CSIC agreement with Springer Nature. This funded was FRQNT team research project 286905, Fonds de recherche du Québec – Nature et technologies, 286905, 286905.

Data availability Data and code are available in GitHub at <https://github.com/quimcanelles/SBW-MGMT>.

Declarations

Conflict of interest The authors declare no conflicts of financial interest.

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